

## Hard-coded (if-condition based & manual calculation) Alerting:

After clicking “Show Results”, compute the following statements then show their results in a concise, readable and modern way. These results must be shown under the “Probabilities” table (above “Deeper AI Analysis” button). Create a new section “Alerts” and show these results inside it.

We will have 3 final results (Porosity, permeability, and Injectivity)  
Follow the best practice.

PETE (Injection & water section in the app):

1- Porosity value (porosity\_fraction in the csv file):

If Porosity value < 10 % = porosity is low

If Porosity value 10 - 20 % = porosity is good

If Porosity value > 20% porosity is Excellent

2- permeability value (permeability\_md in the csv file):

If permeability value < 10 mD = permeability is low

If permeability value 10 - 100 mD permeability is good

If permeability value > 100 mD permeability is excellent

3- Injectivity index:

Injectivity index =  $\text{injection\_rate\_bwpc} / (\text{required\_injection\_pressure\_psi} - \text{reservoir\_pressure\_psi})$

We do not have Injectivity\_index\_bwpc\_per\_psi directly in the current inputs fields. But you have to calculate it using the above formula. Injection\_rate\_bwpc, required\_injection\_pressure\_psi, and reservoir\_pressure\_psi are provided by the user.

After calculating Injectivity index, use the following if-condition to get the result:

If injectivity index value < 10 = injectivity value is poor

If injectivity index value from 20 to 40 = injectivity value is good

If injectivity index value > 50 = injectivity index is excellent